

Operating Systems

Software backbone of a computer

Learning Objectives

1

Understand
concept of
Operating
system (OS).

2

Differentiate
between
different OSs.

3

Witness the
functions of
OS.

4

Experience the
need of OS in
daily life.

Operating System



The **Operating System (OS)** sits **between** hardware and application programs.



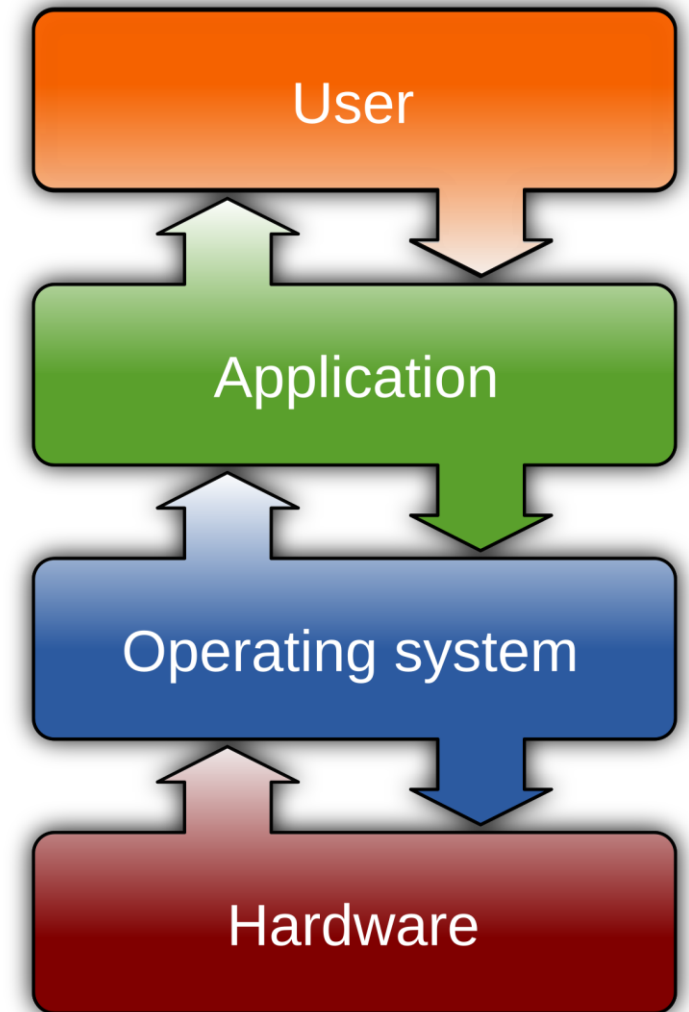
Hardware: CPU, memory, storage, and I/O devices form the foundation.



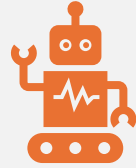
Application Programs: Software like word processors, browsers, and games run on top.



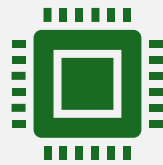
OS Role: Acts as an **interface** between applications and hardware and manages **resources** and provides services to applications.



Examples



Operating Systems:
Windows, Android, iOS



Hardware: Snapdragon
(Qualcomm), Apple M-series
(M1/M2/M3), NVIDIA GeForce



Application Programs:
Instagram, Snapchat,
WhatsApp

Human analogy



Soul sits between body (physical organs) and behavior/thoughts (actions, decisions).



Soul: The conscious controller responsible for awareness and decisions.



Body: The physical structure that executes actions.



Behavior/Thoughts: The observable actions and mental outcomes of decisions.

Can you list the functions that OS do for you?



Functions of an Operating System



Process Management: Controls execution of programs and multitasking.



Memory Management: Keeps track of memory usage and allocation.



File Management: Organizes files and folders for storage and access.



Device Management: Manages input/output devices like keyboard, printer, and storage.



Security & Access Control: Protects data and prevents unauthorized access.



User Interface: Provides easy interaction through GUI or command line.

Examples of Popular Operating Systems

Windows: Widely used on personal computers.

Linux: Open-source, used on servers and desktops.

macOS: Used on Apple computers.

Android: Mobile OS for smartphones and tablets.

iOS: Mobile OS for Apple devices (iPhone, iPad).

How operating system
helps you in your daily
life?

Simple OS Examples in Daily Life



Using Windows to write a document



Using Android to install an app



Using Linux for server tasks

Summary

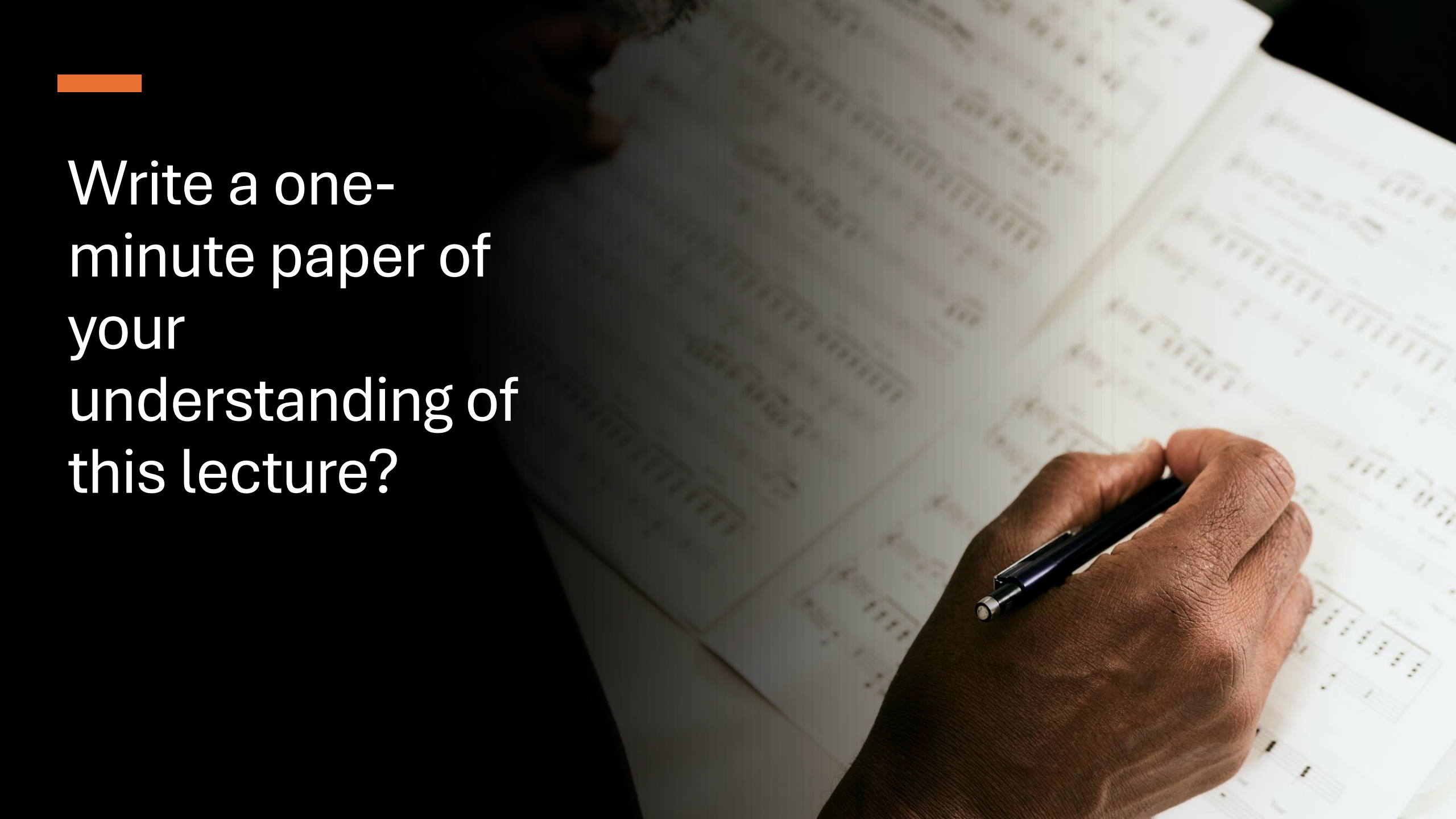
Acts like a **manager** between apps and device hardware

Lets apps like Instagram, games, and browsers run smoothly

Manages phone/laptop resources (battery, RAM, storage)

Handles touch, camera, mic, and internet automatically

Keeps the device secure and responsive

A close-up photograph of a person's hand holding a dark blue pen, poised to write on a document. The document contains several tables with multiple columns and rows of text and numbers. The lighting is soft, highlighting the texture of the paper and the skin of the hand. In the top left corner, there is a small orange rectangular graphic element.

Write a one-minute paper of your understanding of this lecture?